NEXUS LEARNING MANAGEMENT SYSTEM

Enterprise Architecture, Feature Showcase & Complete Technical Defense Guide

Platform: LeadGroup Nexus LMS v2.0	**Stack:** FastAPI • SQLite WAL • Jinja2	**Date:** November 2026	**QA Score:** 90/90 PASSED (100%)

1. Executive Summary & Value Proposition

Nexus LMS is a unified, high-concurrency learning management system developed to serve both institutional academic programs and rigorous competitive test preparation (such as UPSC Civil Services Examination). Designed with a zero-client-build philosophy using asynchronous FastAPI, Server-Side Rendered Jinja2, and a Write-Ahead Logged SQLite database, it eliminates front-end lag, provides instant page painting, and enforces strict security and role-based data isolation across multiple academic departments.

Core Innovations Delivered:

• **Full UPSC Civil Services Preparation Ecosystem:** 2 complete subjects (Indian Polity & Governance, Indian History & Culture), 8 modules, 24 YouTube video lectures, 40 PDF study notes, 40 functional quizzes with 400 Prelims questions, and 320+ enrolled candidates.

• **Subjective UPSC Mains Answer Writing:** End-to-end PDF answer sheet submission, instant mentor notifications, grading rubric scoring (Max 20 marks), and structured mentor feedback critiques.

• **Department-Level Feature Gating:** Granular toggle button allowing administrators to activate/deactivate Answer Writing on a per-department basis, dynamically updating student course players, teacher evaluation queues, and course editors.

• **Personalized 1-on-1 Mentoring System:** Two-way scheduling engine supporting student-initiated requests and faculty-initiated targeted interventions.

• **Authentic Certificate Issuance & QR Verification:** Cryptographic QR code generation and public instant verification portal (`/verify/{id}`).

2. System Architecture & Component Mindmap

The flowchart below illustrates the architectural tiers of Nexus LMS, from public ingress to persistent storage and generative AI.

3. Technology Stack & Architectural Decisions

Layer	Technology	Version / Spec	Technical Rationale / Production Benefits
Backend Framework	FastAPI	Python 3.9+ / Async	Asynchronous ASGI throughput, auto-generated OpenAPI documentation, Pydantic type validation.
Web Server	Uvicorn	ASGI Standard	Non-blocking event loop handling hundreds of student requests concurrently with minimal RAM.
Database Engine	SQLite 3 (WAL)	Write-Ahead Logging	Zero-config portability, high concurrent read throughput, crash safety, and single-file backup ease.
Authentication	JWT + Bcrypt	HS256 in HttpOnly	Stateless session scaling; immune to browser XSS token theft; salted bcrypt password hashing.
Presentation	Jinja2 + Tailwind	SSR + Tailwind 3.4	Instant first-contentful paint, zero client bundler overhead (no npm/webpack runtime), responsive UI.
Generative AI	Google Gemini	gemini-2.5-flash	Low-latency structured AI output for quiz mistake explanation and assessment authoring.
Public Gateway	Cloudflare Tunnel	Argo Tunnel Service	Encrypted global HTTPS ingress without port forwarding or exposing local firewall ports.

4. Role-Based Access Control (RBAC) Matrix

Nexus LMS enforces four primary user roles with strict server-side boundary checks and redirection middleware:

Permission / Feature Capability	Student	Faculty / Teacher	Department Admin	Viewer
Browse Enrolled Curriculum & Stream Videos	**YES**	**YES**	**YES**	**YES**
Download PDF Study Materials & Notes	**YES**	**YES**	**YES**	**YES**
Attempt Timed Prelims Quizzes with Instant Scoring	**YES**	NO	NO	NO
Submit Handwritten UPSC Mains PDF Answer Copies	**YES**	NO	NO	NO
Grade Papers, Award Marks & Write Critique Notes	NO	**YES**	**YES**	NO
Request 1-on-1 Guidance Session	**YES**	NO	NO	NO
Initiate Direct 1-on-1 Session with Student	NO	**YES**	**YES**	NO
Assign Teachers to Pending Mentoring Requests	NO	NO	**YES**	NO
Author / Edit Curriculum, Lessons & Quizzes	NO	**YES (Assigned)**	**YES (Global)**	NO
Toggle Department Answer Writing Feature	NO	NO	**YES**	NO
Onboard Students via Single Form or Bulk CSV	NO	NO	**YES**	NO
Transfer Students Between Department Buckets	NO	NO	**YES**	NO
Schedule Live Masterclasses & Virtual Halls	NO	**YES**	**YES**	NO
Issue Official Authenticated Completion Certificates	NO	NO	**YES**	NO

5. Multi-Tenant Department Bucketing & Answer Writing Gating

To serve diverse disciplines (e.g. Foreign Languages vs. UPSC Civil Services), the platform organizes learners, faculty, and content into **Department Buckets**. Each bucket can independently toggle subjective features off or on.

Dept ID	Department Name	Code	Ans Writing	Target Curriculum	Faculty In-Charge
1	**Foreign Language**	FOREIGN_LANG	**DISABLED (0)**	German A1, French Conversational Mastery, Spanish	Prof. Rameez (T-ALAN)
2	**Test Prep**	TEST_PREP	**ENABLED (1)**	UPSC Indian Polity, Indian History, IIT-JEE Foundation	Dr. A. K. Sharma (T-UPSC-01)

321	**UPSC Coaching**	UPSC_ACAD	**ENABLED (1)**	Civil Services Advanced Prelims & Mains Program	Dr. A. K. Sharma (T-UPSC-01)

Cascading Effects of the Toggle:
• **Student Course Player (`course_player.html`):** If enabled, the 'UPSC Mains Answer Copy Submissions' section and upload PDF modal are rendered. If disabled, they are cleanly removed from the DOM.
• **Upload Endpoint Security:** `POST /api/course/{id}/upload-answer-sheet` strictly checks `is_ans_writing_enabled_for_course()`. Direct unauthorized API requests are blocked with **HTTP 403 Forbidden**.
• **Teacher Dashboard:** The 'Pending Answer Sheet Evaluations' queue is shown exclusively to faculty whose department requires subjective evaluation.
• **Course Editor:** The Mains Evaluation grading panel is dynamically shown only on courses hosted in enabled departments.

6. UPSC Civil Services Examination Preparation Ecosystem

The platform is pre-loaded with a complete, operational UPSC test preparation structure covering General Studies Papers 1 and 2:

Subject & Code	Module Name	Video Lectures	PDF Notes	Quizzes	Prelims MCQs
Indian Polity & Governance (UPSC-GS2-POLITY)	M1: Historical Background & Making of Constitution	3 Lectures	5 Docs	5 Quizzes	**50 Questions**
	M2: Fundamental Rights & Directive Principles	3 Lectures	5 Docs	5 Quizzes	**50 Questions**
	M3: Union and State Government	3 Lectures	5 Docs	5 Quizzes	**50 Questions**
	M4: Constitutional & Statutory Bodies	3 Lectures	5 Docs	5 Quizzes	**50 Questions**
Indian History & Culture (UPSC-GS1-HISTORY)	M1: Ancient Indian History	3 Lectures	5 Docs	5 Quizzes	**50 Questions**
	M2: Medieval Indian History	3 Lectures	5 Docs	5 Quizzes	**50 Questions**
	M3: Modern Indian History	3 Lectures	5 Docs	5 Quizzes	**50 Questions**
	M4: Indian National Movement	3 Lectures	5 Docs	5 Quizzes	**50 Questions**
TOTALS	**8 Modules Across 2 UPSC Subjects**	**24 Videos**	**40 PDFs**	**40 Quizzes**	**400 MCQs**

7. Subjective Mains Answer Writing & Evaluation Workflow

In addition to objective MCQ tests, Civil Services preparation demands subjective answer critique. The platform provides a full evaluation lifecycle for handwritten answer scripts:

Step	Evaluation Stage	Actor & Workflow Description	Platform Action & Status
1	**Candidate Submission**	Aspirant selects topic/module reference and uploads handwritten PDF answer sheet in Course Player.	File stored securely; status initialized to pending_review.
2	**Faculty Notification**	System dispatches real-time notification alert to subject faculty with direct evaluation link.	Faculty evaluation queue increments with new submission.
3	**Evaluation & Grading**	Faculty inspects PDF in-app, enters numerical marks (e.g. 15.5 / 20.0), and writes critique feedback.	Status transitions to evaluated; audit log recorded.
4	**Feedback & Review**	Student inspects evaluated answer copy with marks, breakdown, and mentor critique comments.	Candidate receives feedback alert; marks recorded in profile.

8. Personalized 1-on-1 Mentoring System

Aspirants frequently require personalized strategy counseling. The system features two integrated mentoring pathways:

Workflow Type	Initiator	Intermediary / Actions	Session Resolution

Workflow A: Student-Initiated Request	**Student Aspirant**	• Student submits topic & preferred dates • Admin inspects queue on `/department#tab-mentoring` • Admin allocates faculty member & meeting link (Google Meet)	Teacher conducts session and records permanent post-session advice notes in student file.
Workflow B: Teacher-Initiated Invite	**Subject Faculty**	• Teacher targets student needing intervention • Teacher proposes topic, date, and time slot • Student receives dashboard notification to confirm or suggest revised time	Session status transitions to `scheduled` then `completed` with structured takeaway points.

9. Certificate Issuance & QR Verification Gateway

When students complete their curriculum, the administrator can generate an authentic completion certificate. Each certificate receives a unique immutable identifier (e.g. `LBPL-2026-000008`) and embeds a high-resolution QR code that can be scanned by employers or verifiers to load the public authenticity gateway (`/verify/{id}`) without requiring login.

Component	Implementation Specification	Security & Fraud Prevention Mechanism
Certificate Identifier	Structured serial: `LBPL-YYYY-SEQUENCE` (e.g. `LBPL-2026-000008`)	Indexed unique constraint in SQLite database; prevents tampering or collision.
Embedded QR Code	RFC-standard Base64 PNG Data URI generated via Python `qrcode`	Embedded directly in the HTML/PDF; requires no external image CDN dependencies.
Public Verification	Endpoint: `/verify/{certificate_id}` (Publicly reachable)	Live verification badge confirming recipient name, course, completion date, and issuing authority.
Lifecycle Operations	Issue • Revoke • Re-issue with audit lineage	Revoked certificates display prominent red warning banners explaining grounds for cancellation.

10. Complete Database Schema Dictionary

The platform persists state in a relational SQLite 3 database operating under Write-Ahead Logging (`PRAGMA journal_mode = WAL;`).

Table Name	Records Count	Entity Purpose	Key Relational Columns & Foreign Keys
`users`	332	Accounts (Admin, Teachers, Students)	`id`, `user_id` (Unique), `full_name`, `email`, `role`, `department`, `password_hash`
`departments`	3	Multi-tenant academic buckets	`id`, `name`, `code`, `description`, `ans_writing_enabled`
`courses`	8	Course catalog and subjects	`id`, `title`, `code`, `department`, `teacher_id` (FK → users.id), `is_published`
`modules`	18	Course curriculum chapters	`id`, `course_id` (FK → courses.id), `title`, `position`
`lessons`	223	Videos, PDF notes, and quiz anchors	`id`, `module_id` (FK → modules.id), `title`, `content_type`, `content_value`
`quizzes`	149	Assessment metadata and timers	`id`, `lesson_id` (FK → lessons.id), `title`, `passing_score`, `time_limit_minutes`
`quiz_questions`	739	MCQs with options A-D and rationale	`id`, `quiz_id` (FK → quizzes.id), `question_text`, `correct_option`, `explanation`
`quiz_attempts`	1,551	Student exam submissions & scores	`id`, `quiz_id`, `user_id`, `score`, `score_percent`, `passed`, `submitted_at`
`enrollments`	644	Student course registrations	`id`, `user_id` (FK → users.id), `course_id` (FK → courses.id), `enrolled_at`
`lesson_progress`	3,967	Student progress tracking	`id`, `user_id`, `lesson_id`, `completed_at`
`answer_sheet_submissions`	19	Subjective Mains PDF papers & marks	`id`, `user_id`, `course_id`, `file_path`, `marks_awarded`, `feedback`, `status`
`mentoring_sessions`	6	1-on-1 personalized guidance logs	`id`, `student_id`, `teacher_id`, `topic`, `status`, `scheduled_date`, `meeting_link`
`live_classes`	7	Virtual hall broadcast sessions	`id`, `course_id`, `teacher_id`, `title`, `meeting_url`, `scheduled_at`

`certificates`	6	Verifiable completion certificates	`id`, `certificate_id` (Unique), `student_id`, `course_id`, `qr_code_data_uri`
`notifications`	135	System alerts and bell dispatch	`id`, `user_id`, `title`, `message`, `link_url`, `category`, `is_read`
`audit_logs`	740	Administrator compliance ledger	`id`, `user_id`, `action`, `resource_type`, `resource_id`, `details`, `created_at`

11. Comprehensive Quality Assurance Audit (100% Pass)

An automated test runner (`test_comprehensive_qa.py`) validates every system function and security boundary across 10 functional suites:

Suite	Functional Domain Under Test	Test Scenarios	Passed	Failed	Verdict
Suite 1	Authentication, Salted Hashes, JWT Cookies & Strict RBAC Portal Isolation	14	14	0	**100% PASS**
Suite 2	Department Bucketing, Multi-Tenancy & Answer Writing Toggle Flag	9	9	0	**100% PASS**
Suite 3	Teacher Studio, Course Editor, Video Parsing & PDF Linking	11	11	0	**100% PASS**
Suite 4	Student Directory, Unicode NFKD Normalization & Duplicate Detection	8	8	0	**100% PASS**
Suite 5	Course Enrollment, Dual-ID Resolution, Cross-Dept Access & Unassign	8	8	0	**100% PASS**
Suite 6	Student Course Player, YouTube Streaming, PDF Viewer & Progress Bars	8	8	0	**100% PASS**
Suite 7	Timed Prelims MCQ Engine, Scoring Algorithms & Attempt History	7	7	0	**100% PASS**
Suite 8	UPSC Mains Answer Sheet Submissions, 403 Security Gating & Teacher Grading	9	9	0	**100% PASS**
Suite 9	1-on-1 Mentoring Workflows (Student-Initiated & Teacher-Initiated)	7	7	0	**100% PASS**
Suite 10	Live Virtual Masterclasses, Certificate QR Generation & Public Verification	8	8	0	**100% PASS**
TOTAL	**Comprehensive End-to-End System Verification**	**90**	**90**	**0**	**100.0% PASS**

12. Live Access & Demonstration Directory

Active Public URL (Worldwide Access): https://council-district-stake-python.trycloudflare.com
Localhost Development URL: http://localhost:8000 or http://127.0.0.1:8000

Role / Profile	Login ID	Password	Portal Entry Point	Key Features to Demonstrate
Department Admin	**`DEPT-HEAD` *(or `admin`)***	`admin123`	`/department`	Department Buckets, Ans Writing Toggle Switch, Student Pool, Mentoring Queue, Certificate Engine.
UPSC Faculty	**`T-UPSC-01`**	`password123`	`/teacher`	UPSC Mains Pending Evaluation Queue, Course Editor for Polity & History, 1-on-1 Invite Modal.
Language Faculty	**`T-ALAN` *(or `teacher`)***	`teacher123`	`/teacher`	Foreign Language Studio, Module & Lesson Editor, Live Class Scheduler.
UPSC Aspirant	**`UPSC-2026-0102`**	`password123`	`/student`	Course Player for Polity/History, Answer Copy PDF Upload, Interactive Timed Quizzes, Mentoring Request.
Language Student	**`CS2026-002` *(or `student`)***	`welcome123`	`/student`	Cross-enrolled student profile, active course badges, language player (Answer Writing cleanly omitted).